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Livio Rating (LR) — Corporate Credit Risk Methodology

Proposed institutional-grade framework · Version 2.0 · 19 October 2026

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Overview

The methodology should go substantially beyond conventional financial ratio analysis. The objective is to assess not only whether a company can repay its debt under current conditions, but whether it can continue to service and repay that debt when its business model, management, supply chain, financing or operating environment is disrupted.

I would build the methodology around three fundamental questions:

1. Financial resilience: Can the company service and repay its obligations based on its current financial position and expected cash generation?
2. Structural resilience: Will the company remain competitive and economically viable over the life of the bond?
3. Shock resilience: Can the company survive plausible adverse events, including geopolitical conflict, financial crises, commodity shocks, technological disruption and management failures?

The important distinction is between risk exposure, vulnerability and actual financial impact. For example, an increase in oil prices can be harmful to an airline, beneficial to an oil producer and relatively neutral to a software company. The methodology must capture these differences quantitatively wherever possible.

A credible LR framework should therefore combine financial metrics, forward-looking business analysis, event-risk assessment and explicit stress testing, with a transparent separation between observable facts and analytical judgments.

1. Overall methodology architecture

The published framework should distinguish between the base credit score, the financial consequences of adverse scenarios, and exceptional credit events.

A. Base credit assessment LR Base · 0–100
Financial strength + business resilience + management + structural exposures

B. Scenario engine
Recalculate cash flows, liquidity and leverage under defined shocks

C. Event-risk overlay
Governance failures, litigation, extraordinary transactions and other discrete events

D. Final corporate credit assessment LR Final · 0–100
With separate scenario results, confidence indicators and critical-risk flags

This is directionally consistent with established rating-agency practice, which distinguishes business and financial risk, modifiers and group support, evaluates financial policy and event risk, and separates foreseeable vulnerabilities from unforeseen event risks.

LR's proposed differentiation: a standardized, transparent, issuer-specific shock-transmission model that makes the consequences of external risks visible to bond investors rather than relying only on a broad qualitative risk judgment.

2. Base credit assessment: 100 points

I recommend the following starting allocation for nonfinancial corporates.

A. Financial strength **35%**

Leverage 9%; interest coverage 6%; FFO/debt 6%; FCF generation 6%; profitability and volatility 4%; leverage trajectory 4%

B. Liquidity and repayment **20%**

24-month liquidity 7%; maturity profile 5%; refinancing flexibility 4%; covenants and contingent obligations 4%

C. Business model resilience **20%**

Competitive position 5%; industry outlook 4%; customer/product concentration 3%; technological and China-related disruption 4%; geographic and supplier diversification 4%

D. Management and governance **10%**

Governance 3%; financial policy 3%; key-person dependency and succession 2%; execution and risk management 2%

E. Structural and external exposure **15%**

Geopolitical 4%; commodities and energy 3%; financial-system sensitivity 3%; physical and climate exposure 2%; legal/regulatory 3%

These weights are proposed research hypotheses, not empirically established optimal weights. Before publication as a validated rating methodology, they must be tested against historical defaults, rating migrations and observed financial deterioration.

A crucial rule: factors in Pillar E measure persistent exposure and vulnerability. The scenario engine measures the incremental consequences of shocks. We must not deduct the same risk twice.

3. Comprehensive corporate risk universe

The following taxonomy is designed as the master risk register. Each risk receives a materiality assessment, evidence record and defined transmission path into financial performance.

A. Geopolitical, conflict and international trade

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Interstate war	Operations, assets, employees or customers in conflict zones	Production losses, asset impairment
Regional military escalation	Exposure to Taiwan Strait, Middle East, Eastern Europe and other flashpoints	Supply interruption, logistics and demand
Trade wars	Tariff exposure by product and destination	Margin compression, lost volumes
China-driven disintermediation	Ability of Chinese manufacturers to replace the issuer's products or customer relationships	Market-share loss, lower prices
Chinese technology competition	Relative cost, quality, scale and innovation	Structural EBITDA decline
Sanctions	Counterparties, countries, technologies and currencies	Inability to trade, stranded assets
Export controls	Dependence on restricted chips, software or equipment	Production and revenue disruption
Nationalization/expropriation	Physical assets in higher-risk jurisdictions	Asset loss
Political instability	Civil unrest, regime change, institutional deterioration	Operational shutdown, higher costs
Strategic shipping chokepoints	Taiwan Strait, Strait of Hormuz, Suez, Panama	Freight, fuel and inventory costs
Capital and currency controls	Ability to repatriate cash or service foreign debt	Debt-payment constraints
Protectionism/localization	Requirements for local manufacturing or procurement	Additional capex, reduced profitability

China risk must be measured at the product and market level, not assigned mechanically based on the issuer's nationality. For example, a European industrial company could have limited sales in China but substantial exposure to Chinese competitors in Europe. Its competitive risk could therefore be high despite low direct Chinese revenue.

B. Economic, financial and monetary risks

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Global recession	GDP sensitivity of demand	Revenue and EBITDA decline
Financial crisis	Funding availability and credit spreads	Refinancing failure
Banking-system crisis	Deposit concentration and reliance on bank facilities	Liquidity constraints

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Interest-rate increases	Floating-rate debt, refinancing calendar	Higher interest expense
Inflation	Ability to pass through labor and input costs	Margin deterioration
Deflation	Pricing power and fixed debt obligations	Falling cash generation
Currency depreciation	Currency mismatch between cash flows and debt	Higher debt-service burden
Sovereign default	Exposure to sovereign counterparties, banking and capital controls	Receivables losses, financing disruption
Commercial real-estate crisis	Tenant, collateral and customer exposures	Impairment, demand decline
Credit-market closure	Debt maturities during inaccessible markets	Funding gap
Counterparty default	Concentrated receivables and hedging counterparties	Cash losses
Pension funding crisis	Defined-benefit deficits and mandatory contributions	Increased cash obligations

C. Commodities, energy and natural resources

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Oil price spike	Fuel intensity, hedging, price pass-through	Higher costs or revenue
Oil price collapse	Upstream production and energy-linked demand	Lower revenue
Natural gas disruption	Gas dependence, contracted supply	Higher costs, shutdown
Electricity price surge	Power intensity and electricity contracts	EBITDA reduction
Critical mineral shortages	Copper, lithium, rare earths and other essential inputs	Production constraints
Agricultural commodity shocks	Food-processing and agricultural exposure	Input inflation
Water scarcity	Water-intensive facilities and geographic exposure	Lower capacity, capex
Carbon pricing	Emissions intensity and regulatory exposure	Higher compliance costs
Energy transition	Stranded assets and changes in end-market demand	Long-term earnings erosion

Every commodity shock must be assessed for its direction, magnitude, hedging protection, pass-through capacity and time lag.

D. Technology and business-model disruption

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
AI substitution	Proportion of products/services that AI could replace	Structural revenue loss
Automation	Competitors' ability to reduce costs faster	Margin pressure

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Platform disintermediation	Dependence on distributors, marketplaces and aggregators	Loss of customer ownership
Technological obsolescence	Product life cycle, R&D effectiveness	Market-share loss
Cyberattack/ransomware	Cybersecurity controls and recovery capabilities	Shutdown, recovery cost
Cloud concentration	Dependency on one cloud or software provider	Business interruption
Data breach	Sensitive data, compliance and liability	Fines, litigation, customer losses
Patent expiry	Revenue protected by intellectual property	Pricing collapse
Product substitution	Alternative technologies or materials	Declining addressable market
Digital infrastructure failure	Dependency on payments, connectivity and digital systems	Revenue interruption

E. Supply chain and operational continuity

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Single-source suppliers	Availability of qualified substitutes	Production stoppage
Geographic concentration	Production clustered in one country or region	Correlated disruption
Semiconductor shortages	Critical electronic components	Production reduction
Logistics disruption	Dependence on shipping lanes and transport modes	Delays, increased costs
Supplier bankruptcy	Financial strength of critical suppliers	Replacement costs
Manufacturing accidents	Safety and redundancy of major facilities	Shutdown, liabilities
Quality failure/product recall	Product safety, traceability and recall exposure	Cash outflow, brand damage
Labor strikes	Union concentration, labor relations	Lost production
Skilled labor shortages	Availability of essential technical capabilities	Capacity and cost pressures
Infrastructure failures	Grid, water, telecom and transport dependencies	Operating interruption
Inventory disruption	Just-in-time exposure, safety stock	Working-capital shock

F. Health, environmental and natural-catastrophe risks

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Global pandemic	Business shutdown sensitivity and demand shifts	Revenue, labor and liquidity
Regional epidemic	Geographic concentration and workforce exposure	Local interruption
Earthquakes	Facilities in seismic zones	Physical losses
Flooding	Asset locations and business continuity	Damage and downtime
Hurricanes/typhoons	Production, distribution and supplier exposure	Disruption and repair costs
Extreme heat	Labor productivity, cooling and water requirements	Higher operating costs

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Wildfires	Facilities and logistics corridors	Shutdown and asset damage
Drought	Agricultural, energy and industrial dependencies	Supply constraints
Environmental accidents	Pollution and remediation exposure	Fines, cleanup, litigation
Insurance withdrawal	Availability and affordability of catastrophe coverage	Higher retained losses

G. Management, governance and human-capital risks

This category deserves particular attention because conventional financial statements often fail to capture management concentration and operational fragility.

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
CEO death/incapacity	Succession plan and delegation of authority	Strategic and operating disruption
Founder dependency	Critical relationships, knowledge and decisions held by one person	Business continuity risk
CFO departure	Financial controls and funding relationships	Reporting and financing disruption
Key technical expert loss	Proprietary knowledge concentrated in individuals	Product/production interruption
Management-team turnover	Stability of senior leadership	Execution deterioration
Poor succession planning	Identified and prepared replacements	Extended leadership vacancy
Fraud/accounting manipulation	Audit findings, restatements and controls	Liquidity crisis, covenant breach
Aggressive acquisitions	Management's leverage appetite	Debt increase
Shareholder extraction	Dividends and buybacks funded by debt	Reduced creditor protection
Ownership disputes	Shareholder litigation and control conflicts	Strategic paralysis
Poor board oversight	Independence, risk oversight and incentives	Excessive risk-taking
Executive compensation	Incentives favoring short-term equity returns over creditor interests	Higher financial risk
Succession disputes	Competing heirs or controlling shareholders	Governance instability

A key-person event should not automatically cause a major score deduction. A company with a strong succession plan, delegated authority and institutionalized customer relationships may be highly resilient to the loss of an executive. Conversely, a founder-controlled company in which one individual controls technology, funding relationships and customer contracts may have substantial key-person risk.

H. Legal, regulatory and strategic event risks

RISK	WHAT LR MUST INVESTIGATE	FINANCIAL TRANSMISSION
Antitrust enforcement	Market dominance and competition investigations	Fines, divestitures
Major litigation	Probability and magnitude of adverse judgments	Cash outflow
Product liability	Potential claims and insurance protection	Liabilities
Regulatory prohibition	Potential restrictions on core products	Revenue loss
Tax-law changes	Tax jurisdiction and structure	Higher cash taxes
License revocation	Dependence on licenses and concessions	Operating interruption
Debt-funded acquisition	Acquisition size and financing policy	Leverage increase
Leveraged buyout	Change-of-control provisions and owner incentives	Capital structure deterioration
Corporate restructuring	Execution and financing requirements	Costs, covenant pressure
Parent-company distress	Group funding dependence and cash extraction	Liquidity loss
Guarantee withdrawal	Contractual support and guarantee enforceability	Reduced credit protection
Unexpected capital expenditure	Compliance, environmental or operational requirements	FCF deterioration

4. How to convert these risks into a quantitative assessment

A risk checklist is useful, but insufficient. A credible model needs to show how an identified risk can impair the company's capacity to repay debt. I propose a five-stage calculation.

Stage 1 — Exposure. Determine which portion of revenue, EBITDA, assets, procurement, production or debt is exposed to the event.

Stage 2 — Vulnerability. Evaluate the company's ability to mitigate the shock through pricing, hedging, insurance, diversification, substitution or contingency plans.

Stage 3 — Financial impact. Translate the shock into changes in revenue, EBITDA, working capital, capital expenditure, free cash flow, interest expense and debt.

Stage 4 — Timing and recovery. Model the event's duration, timing relative to debt maturities, recovery path and any permanent deterioration.

Stage 5 — Repayment consequences. Recalculate leverage, interest coverage, liquidity and the issuer's ability to meet contractual payments under the stressed scenario.

Illustrative example: oil price shock

PARAMETER (HYPOTHETICAL EUROPEAN INDUSTRIAL COMPANY)	BASE CASE	OIL SHOCK
Revenue	€5,000m	€5,000m
EBITDA	€800m	€680m
Net debt	€2,000m	€2,000m
Annual cash interest	€100m	€100m
Net debt / EBITDA	2.50×	2.94×
EBITDA / interest	8.00×	6.80×

Assume oil rises by 50%, the company experiences €200m in incremental gross operating costs, and it recovers €80m through hedging and customer price increases. EBITDA consequently falls by €120m. The credit assessment must then examine whether the company still generates sufficient free cash flow, whether its debt covenants remain satisfied and whether it can refinance or repay upcoming maturities. The model should also test whether the shock persists for one, two or three years.

5. Scenario engine: the distinctive LR component

I recommend a common set of scenarios applied consistently across issuers, with company-specific transmission assumptions.

SCENARIO	STANDARDIZED SHOCK SPECIFICATION	PRIMARY OUTPUT
Global recession	Revenue volume decline calibrated to industry sensitivity	EBITDA and FCF
Severe financial crisis	Credit spread increase and temporary market-access closure	Refinancing gap
Oil crisis	+50% oil price, with a second more severe sensitivity	Net EBITDA impact
Energy crisis	Material gas/electricity price increase and supply interruption	Operating costs
China competitive shock	Accelerated price and market-share pressure in exposed products	Structural earnings loss
Taiwan conflict	Semiconductor and trade disruption, plus regional demand effects	Production and cash flow
Middle East conflict	Energy and shipping disruption	Costs and logistics
Trade war	Higher tariffs and market-access restrictions	Margins and volumes
Pandemic	Demand interruption, absenteeism and operating restrictions	Liquidity runway
Supply chain failure	Loss of a critical supplier for a defined period	Lost production
Cyberattack	Temporary system outage and recovery expenditure	Cash losses
Interest-rate shock	+300 basis points on relevant debt exposure	Interest coverage
Currency crisis	Material depreciation in exposed currencies	FX losses and debt service
Climate catastrophe	Loss of a major facility and associated insurance recovery	Liquidity and asset losses
Key-person event	Sudden loss of a critical executive or technical leader	Execution and revenue effects
Combined systemic crisis	Recession + financing closure + commodity shock	Maximum credible stress

These shock magnitudes are illustrative scenario design choices, not estimated forecasts or calibrated tail-event quantiles. They should be refined using historical observations, sector characteristics and empirical testing.

Scenario impact should not be an arbitrary score

For each scenario s, calculate LR(s) = f(stressed financial and business inputs), then measure ΔLR(s) = LR(base) − LR(s). The model should report both the score change and the underlying financial deterioration.

Manufacturer: LR 78 → 43 under combined stress.

Utility: LR 74 → 66 under combined stress.

This allows an investor to understand that two companies with identical base credit scores may have substantially different vulnerabilities. A manufacturer with LR 78 that falls to 43 under a plausible combined stress may be less resilient than a utility with LR 74 that remains at 66. However, the combined scenario must explicitly model correlated shocks. It cannot be constructed by simply adding the separate losses from every scenario.

6. Incorporating event likelihood without false precision

Not every possible event should receive an invented annual probability. I recommend three classes:

RISK CLASS	METHODOLOGY	EXAMPLE
Structural risk	Incorporate into base forecast and competitive assessment	Persistent Chinese price competition
Measurable stochastic risk	Use historical distributions and calibrated scenarios	Interest rates, commodity prices
Discrete event risk	Apply conditional scenarios and materiality flags; assign probability only with credible evidence	War, CEO death, nationalization

The important distinction is between a risk that is already present and an event that might occur. For example, reliance on one critical executive is an observable structural vulnerability. The executive's sudden death is a discrete event. The first belongs in the base resilience assessment; the second can be tested as a scenario without inventing a mortality-based credit adjustment.

7. Critical-risk overrides

A weighted-average score has an important weakness: a company could score well on most factors while facing one existential risk. The methodology therefore needs override rules. I propose that the following conditions trigger mandatory review rather than simply reducing the score by a few points:

- Insufficient liquidity to meet contractual obligations within the next 12 months under the base case.
- Material debt covenant breach without a documented waiver or credible remedy.
- A near-term refinancing requirement without demonstrated funding access.
- A credible event threatening the company's ability to continue operating.
- Material financial reporting uncertainty or evidence of serious accounting manipulation.
- A material legal, sanctions or regulatory restriction preventing debt service.

These should trigger a documented credit committee review, possible score cap or distressed-credit classification. The precise caps must be specified and empirically tested before production use. This approach prevents an otherwise strong average score from concealing a critical repayment problem.

8. Model validation and institutional credibility

A comprehensive taxonomy alone will not make LR credible to an experienced rating-agency analyst. The credibility will come from how the model handles evidence, uncertainty, financial causality and historical performance. I recommend six mandatory methodological controls.

CONTROL	REQUIRED STANDARD
Data traceability	Every material input must have a source, observation date, definition and calculation record
Sector comparability	Financial thresholds and risk transmission calibrated by industry
No double counting	Each risk's impact allocated to the appropriate base factor or incremental scenario
Reproducibility	Same verified inputs and methodology version produce the same score
Historical validation	Back-test against defaults, distress events and credit deterioration using information available at each historical date
Uncertainty disclosure	Publish data coverage, confidence, assumptions and unquantifiable risks

Validation tests

Before describing LR as a predictive rating model, I would require:

1. Discrimination: Do lower LR scores precede defaults more frequently than higher scores? Test with ROC-AUC and related ranking metrics.
2. Calibration: If the model eventually estimates default probabilities, do predicted probabilities match observed frequencies?
3. Early-warning performance: Does LR identify deterioration 12–36 months before default or restructuring?
4. Stability: Does the score change for economically meaningful reasons rather than minor reporting differences?
5. Incremental value: Do geopolitical, technology and management-risk factors improve predictive performance beyond conventional financial metrics?
6. Benchmark comparison: How does LR perform relative to published agency ratings, accounting-based models and simpler financial-ratio benchmarks?

The fifth test is especially important. It would demonstrate whether the additional complexity of LR produces genuine analytical value.

9. How we publish this on 10bonds.com

I would position LR as a forward-looking corporate credit resilience methodology, rather than claiming it is more accurate than agency ratings before validation. The public methodology should contain four parts:

PUBLICATION	CONTENT
1. Corporate LR Methodology	Objectives, 100-point framework, all factors, definitions, formulas and scoring rules
2. Global Risk and Scenario Manual	Full taxonomy, shock assumptions, transmission formulas, scenario selection and correlation treatment
3. Data and Model Governance	Sources, missing data, confidence, updates, version control, overrides and validation
4. Worked Corporate Credit Assessment	One complete issuer example with historical inputs, base score, stress scenarios, repayment analysis and audit trail

All published scores should also distinguish the issuer's creditworthiness from the specific bond's seniority, guarantees, collateral, contractual maturity and recovery prospects.

For methodological reference, the following published frameworks are particularly relevant:

- Published corporate rating methodologies of the major credit rating agencies
- Their published management and governance criteria

These are methodological references, not endorsements of LR or evidence that the proposed LR weights have been validated.

10. The key design decision

I would not publish a final corporate LR formula yet. We now have a substantially more complete risk universe, but three technical questions must be resolved before freezing the methodology:

- Weights: Which risks belong in the 100-point base score, and which should affect only stressed outcomes?
- Quantification: How do we convert structural disruption, geopolitical exposure and key-person dependency into repeatable, evidence-based assessments?
- Aggregation: Should the final LR score reflect expected outcomes, downside resilience or a combination, and how do we avoid double-counting correlated risks?

My proposed solution is to preserve LR Base as a transparent 0–100 credit-strength score, publish LR Stress as a separate resilience profile, and develop a final composite only after historical calibration establishes defensible aggregation rules.

That separation would make the methodology more credible than attaching arbitrary numerical penalties to every conceivable event. It would also give 10bonds a useful investor-facing distinction: a company's current financial strength versus its ability to survive future disruptions.

The ambition should be an auditable, forward-looking framework whose added risk factors demonstrably improve credit assessment—not simply a longer checklist than the rating agencies use.